

Monetary Economics and Policy

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Chapter 14: The Future of
Monetary Policy



Outline

- 1 A Benchmark: Neutrality and the “Virtual Room”
- 2 History and the Temptation to Manipulate
- 3 Cryptocurrencies: Competition and Verification
- 4 Implications for the Future Design of Monetary Policy
- 5 Bibliography

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A Stable Monetary Framework

- ▶ The book highlights a complex governance of currency: value anchoring, liquidity provision, crisis management, and fiscal interactions.
- ▶ Because the same institution can stabilize some margins while distorting others, sweeping conclusions can lead to contradictions.
- ▶ Still, offering a unified framework in the book invites a preliminary “big-picture” discussion of the characteristics of a **stable monetary framework** (cf. Friedman 1960, Friedman and Schwartz 1986).

From Barter to the Need for a Monetary System

- ▶ Barter economies are limited by the **double coincidence of wants**.
- ▶ Exchange is confined to proximity and bilateral matching; economic development is constrained.
- ▶ A currency system and market economy can be interpreted as institutions that approximate a more complete coordination device than barter.

Medieval Europe and “Near-Barter” Conditions

- ▶ Cipolla (1956): In medieval Europe the demand for coins was depressed. Coins were only one of many means of settlement.
- ▶ Even debts stipulated in coin units were often settled with commodities (vases, cloths, horses).
- ▶ A central reason was an **inefficient market**: people could not rely on markets to have what they wanted (Cipolla 1956).
- ▶ With weak market reliability, there is little incentive to hold coins for future transactions with third parties.

The “Virtual Room” Benchmark

- ▶ Imagine a hypothetical “virtual room” where all current and future generations reveal their plans.
- ▶ A virtual Walrasian auctioneer adjusts intratemporal and intertemporal prices so plans are mutually compatible.
- ▶ In that ideal benchmark, no currency is needed as an intermediary.
- ▶ The benchmark is useful because it highlights a desideratum: a currency should be as **neutral** as possible relative to the allocation that would obtain under full-information coordination.

Currency Irrelevance as a Desideratum

- ▶ If currency were “irrelevant” in the benchmark sense, it would not distort the compatible plans that solve the virtual-room problem.
- ▶ It is hard to imagine a world where a worthless currency improves everyone’s real plans: the benchmark suggests that distortions from currency manipulation are costs, not benefits.
- ▶ In a second-best world, a currency system and market economy may be the best approximation we can get to the virtual-room coordination device.

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The First Distortion: Granting Favored Status

- ▶ Moving away from barter by selecting a commodity as a privileged medium of exchange creates a distortion:
 - ▶ the privileged commodity gains purchasing power relative to the pre-currency allocation,
 - ▶ agents holding more of it benefit disproportionately.
- ▶ Even if it solves the coincidence-of-wants problem, it creates distributional consequences.

Manipulation Under Metallic Coin Regimes

- ▶ Historical practice under sovereign control of coinage:
 - ▶ seigniorage extraction,
 - ▶ debasement of intrinsic value,
 - ▶ monopolistic management of issuance and verification.
- ▶ These practices aimed at appropriating real resources and thus increased distortions.

Inconvertible Paper Currency and Stronger Temptations

- ▶ Under inconvertible paper currency, currency is “what can be printed at will.”
- ▶ This heightens the temptation to manipulate real allocations through monetary expansion.
- ▶ Central banks historically emerged from urgent needs of the state:
 - ▶ financing the state,
 - ▶ averting banking liquidity crises.
- ▶ These actions, one way or another, distort allocations in favor of the state and/or the banking sector.

2007–2008 and COVID: Buyers of Last Resort

- ▶ Recent history: central banks became buyers of last resort for “almost everything” during:
 - ▶ the Global Financial Crisis (2007–2008),
 - ▶ the COVID-19 pandemic.
- ▶ Policymakers faced hard governance decisions; the sequence around Lehman highlights the difficulty:
 - ▶ failure allowed on Sept. 15, 2008,
 - ▶ immediate shift toward broad purchases the next day.
- ▶ Whether such interventions represent “appropriate” money governance is debatable.

Europe: A New Era of Central Bank Backing

- ▶ The Euro Area illustrates evolving monetary–fiscal dynamics:
 - ▶ “whatever it takes” (Draghi, July 26, 2012),
 - ▶ pandemic-era shifts and large-scale national debt purchases.
- ▶ These interventions loosened sustainability constraints and enabled large fiscal actions.
- ▶ The resulting allocation distortions have arguably been reflected in the post-COVID inflation surge.

The Difficult Second-Best Reality

- ▶ From the virtual-room benchmark perspective, achieving a neutral currency is challenging even in a second-best world.
- ▶ Citizens appear “consigned” to currency jurisdictions vulnerable to manipulation.
- ▶ This motivates interest in mechanisms that could discipline or replace national monopoly currency provision.

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Cryptocurrencies as a Watershed Event

- ▶ The emergence of private digital currencies introduces:
 - ▶ competition in currency adoption and usage,
 - ▶ systems that transcend national borders,
 - ▶ the possibility of switching currency systems without relocating.
- ▶ This can be a powerful force for better currencies or at least meaningful choice.

Historical Note: Currency Competition Is Not New

- ▶ A single national currency is relatively recent.
- ▶ In metallic systems, competition among coins with different intrinsic content was common.
- ▶ Such competition often produced Gresham's-law phenomena: “good” coins hoarded, “bad” coins circulate.
- ▶ Modern crypto competition differs by operating globally and digitally.

Hayek: Denationalization and Competition as Discipline

- ▶ Hayek (1976): competition could constrain issuers more effectively than redeemability obligations.
- ▶ He envisions dominance of a currency that maintains purchasing power (a “real dollar”).
- ▶ Different commodity baskets relevant for different groups could imply coexistence of multiple currencies.
- ▶ A stable purchasing-power currency could also be attractive as protection against fiscal and financial dominance.

Friedman and Schwartz (1986): A Cautious View

- ▶ Friedman and Schwartz (1986) take a cautious view: it is **not yet established** that private fiduciary currencies can reliably deliver both efficiency and safety without public authority or a well-defined institutional backstop.
- ▶ They therefore adopt an **agnostic** stance on the optimal monetary arrangement (public vs. private), emphasizing that the answer is ultimately empirical and institutional.
- ▶ Still, allowing **competition** in currency and payments can discipline government issuers: if public management performs poorly, market alternatives can constrain policy mistakes—consistent with their broader theme that **government failure** can be as severe as **market failure** in some circumstances.

Cryptographic Verification: A Second Innovation

- ▶ Cryptocurrencies bring a decentralized verification system based on cryptography.
- ▶ It grants no special privilege except for mathematical consensus.
- ▶ This shifts verification from third-party judgment and courts toward a “cryptography guarantee”.
- ▶ It speaks to contract-enforcement and fraud-prevention difficulties emphasized by Friedman and Schwartz (1986).

Smart Contracts and Decentralized Finance

- ▶ Decentralized finance extends the idea via smart contracts:
 - ▶ automated, rule-based execution,
 - ▶ potentially reduced reliance on subjective enforcement.
- ▶ This technological direction could improve the transparency of risk and the enforceability of promises-to-pay.

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What the Book Helps Us Organize

- ▶ The book provides a framework to reason about:
 - ▶ anchoring the value of an intrinsically worthless currency,
 - ▶ supplying liquidity / safe assets,
 - ▶ macroeconomic stabilization (inflation/output),
 - ▶ crisis intervention and prevention.
- ▶ These dimensions matter regardless of whether the dominant currency is public or private.

Separation of Monetary and Fiscal Policy

- ▶ Chapters 1, 2, and 13 provide arguments for separating monetary and fiscal policy.
- ▶ Giving special guarantees to treasury liabilities or direct deficit finance:
 - ▶ makes currency value depend unnecessarily on fiscal policy,
 - ▶ increases the risk of fiscal dominance and instability.
- ▶ The logic is sharpened by a key insight: the currency does not even require fiscal backing to be anchored (under the principles developed earlier in the book).

Backing Without the Treasury: Assets and Real Backing

- ▶ If seigniorage is weakened by competition, value control hinges more on backing via assets.
- ▶ A central bank can support its liabilities by holding:
 - ▶ high-quality default-free securities,
 - ▶ and/or tangible real assets (e.g. gold).
- ▶ Gold provides an immediate, simple anchor when a newly created currency lacks deep safe-asset markets.
- ▶ Historical note (Cipolla 1956): early acceptance of new gold coins could be difficult, even when intrinsically valuable.

Operational Framework: What to Target

- ▶ Currency-value stability can be encoded as:
 - ▶ inflation targeting,
 - ▶ price-level targeting,
 - ▶ nominal spending targeting.
- ▶ A low positive inflation trend can be useful as a buffer against distortions highlighted in the book.
- ▶ Digitization may reduce the relevance of some distortions (e.g. the ZLB), though other distortions may become more important.

Neutrality as the Issuer's Responsibility

- ▶ In a competitive environment, the issuer of the unit of account should:
 - ▶ assume sole responsibility for stabilizing purchasing power,
 - ▶ play a neutral role, avoiding distortions that favor particular groups.
- ▶ This is the closest feasible approximation to the virtual-room benchmark in a second-best world.

Who Should Supply Liquidity? Lessons from Chapters 4 and 11

- ▶ Chapters 4 and 11 provide contrasting but complementary lessons.
- ▶ **Chapter 4 (frictionless private money creation):**
 - ▶ competition can satisfy liquidity needs,
 - ▶ private liquidity does not interfere with unit-of-account value control.
- ▶ **Chapter 11 (frictions and pseudo-safe assets):**
 - ▶ frictions can produce pseudo-safe liquidity and crises,
 - ▶ poor private liquidity performance can cause contractions and disinflationary pressure,
 - ▶ the issuer must substitute low-quality private liquidity with high-quality public liquidity in crises.

Public Liquidity, Backing, and Dominance Risks

- ▶ One reading of Chapter 11: even beyond crises, the unit-of-account issuer may need to supply liquidity broadly.
- ▶ But abundant liquidity requires equivalent backing:
 - ▶ either high-quality assets on the issuer's balance sheet,
 - ▶ or (if integrated with the treasury) current and future taxes.
- ▶ Both routes create possible dominance risks:
 - ▶ financial dominance through private-sector balance-sheet exposure,
 - ▶ fiscal dominance through treasury linkage.

Why Private Money Creation Breaks Down

- ▶ Chapter 11's crisis mechanism stems from an underdeveloped intermediation process:
 - ▶ opacity of intermediaries' balance sheets (especially asset quality),
 - ▶ limited ability to evaluate risk and liquidity characteristics correctly.
- ▶ New technologies (cryptographic guarantees) could improve transparency and verification:
 - ▶ better pricing of risk and liquidity,
 - ▶ competition can then more reliably determine the amount of liquidity supplied.

Maturity Transformation as a Symptom of Underdevelopment

- ▶ Historically, banks were viewed as money-claim providers through maturity transformation.
- ▶ But...this could be as a symptom of an underdeveloped system rather than a necessary feature.
- ▶ Modern intermediation (e.g. private equity) shows long-term financing can be matched with long-term funding without maturity transformation.
- ▶ A more developed system may separate money and credit markets:
 - ▶ fewer implicit guarantees in private money markets,
 - ▶ reduced encouragement of risk-taking.

Lender of Last Resort: Present vs. Potential Future

- ▶ In today's framework it is hard to dispense with a lender of last resort (e.g. Goodhart 1999).
- ▶ In the more developed framework envisioned here, the need could diminish:
 - ▶ LLR should not address solvency problems,
 - ▶ private equity losses should be borne by equity holders.
- ▶ If money-like claims are appropriately priced and balance sheets are transparent, there is less scope for an LLR role in money markets as well.

A “Stable Monetary Framework” as a Vision

- ▶ A possible stable framework:
 - ▶ the issuer of the unit of account ensures stable purchasing power,
 - ▶ the private sector provides the medium of exchange under competition,
 - ▶ the treasury is one agent among many, without special connection to the unit of account.
- ▶ This is presented as the best approximation to a neutral currency system from the virtual-room perspective.
- ▶ Competition among currencies may allow such a framework to transcend national borders.

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References I

-  Cipolla, Carlo Maria (1956). *Money, Prices, and Civilization in the Mediterranean World*. Princeton: Princeton University Press.
-  Friedman, Milton (1960). *A Program for Monetary Stability*. New York: Fordham University Press.
-  Friedman, Milton and Anna J. Schwartz (1986). “Has Government Any Role in Money?” In: *Journal of Monetary Economics* 17.1, pp. 37–62.
-  Goodhart, Charles (1999). “Myths about the Lender of Last Resort”. In: *International Finance* 2.3, pp. 339–360.
-  Hayek, Friedrich A. (1976). *The Denationalization of Money*. London: Institute of Economic Affairs.